

APT-BENCH: SAMPLING REPORT

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ABSTRACT

This standalone report preserves the supporting subject–cell sampling analyses previously included in the APT-Bench manuscript. These experiments address allocation policies, training-label distribution, and uncertainty, not the primary APT cell-typing benchmark or external disease validation. The report retains the historical analyses and their limitations; separation from the paper does not delete results or promote them to independent claims.

1 SCOPE AND STATISTICAL UNITS

The main APT-Bench paper evaluates APT-only cell typing on new patients. This report instead compares training-allocation policies on APT, COMBAT, and OneK1K. Subject-balanced pooled Macro-F1 normalizes each subject’s confusion matrix by its total cell count before pooling, using a fixed class universe and zero contribution for undefined class F1. Cell-weighted historical results are labeled separately. The report preserves fixed-cohort and fixed-fold limitations, differentiates stability ranges from confidence intervals for mean effects, and does not claim a universal scaling law.

2 SYNTHETIC VALIDATION

2.1 SEQUENTIAL CONTROLS: SYNTHETIC MECHANISM CHECK

The real-data class-matched sampler fixes class quotas and spreads cells across donors within each class. To distinguish these operations, we implement four conditions with common random cell priorities and nested development subject sets: (A) uniform pooled-cell sampling; (B) restriction to the eight-subject reference class set, reserving one cell per class before uniform filling; (C) exact reference class quotas with uniform within-class sampling; and (D) the same quotas with capacity-constrained donor round robin in randomized donor order. At fixed total, exact class proportions and exact counts are equivalent constraints, not separate ablations. This A condition is not the legacy equal-cell-per-subject sampler. Sequential contrasts are order-dependent descriptions, not causal mediation estimates.

We evaluate this implementation on synthetic data, not a semi-synthetic clinical cohort. Each of 20 seeds generates six class centers $\mu_k \sim \mathcal{N}(0, s^2 I_{12})$ and 48 subjects with 400 cells each. Subject label probabilities are $\text{softmax}(\eta_p)$, where $\eta_p \sim \mathcal{N}(0, m^2 I_6)$, and features are $x_{pi} = \mu_{y_{pi}} + b_p + \epsilon_{pi}$, with $b_p \sim \mathcal{N}(0, a^2 I_{12})$ and $\epsilon_{pi} \sim \mathcal{N}(0, I_{12})$ independently. We vary $a, m \in \{0, 1.5\}$ and $s \in \{0.5, 1.5\}$. The first 32 subjects form the development pool; the remaining 16 are untouched test subjects. An unweighted LR probe is refit with training-only standardization for each condition and $P \in \{8, 32\}$ at total 1,600 training cells. All conditions reuse the same realization and test subjects within a seed.

With exchangeable subjects ($a = m = 0$), mean effects are near zero. With label mixture variation alone and $s = 0.5$, the unmatched mean effect is +0.0245, compared with −0.0011 after quota matching. Conversely, with subject offsets alone and $s = 0.5$, quota matching retains a mean effect of +0.0280, compared with unmatched +0.0281. Thus matching does not invariably eliminate a breadth benefit in this generator. Seed variability is substantial; these results do not establish interval coverage, a universal correction, or the mechanism in APT, COMBAT, or OneK1K. Support is complete in these simulations, so they do not test missing-class recovery. Separate real-cohort A–

Offset	Mixture	Separation	A: Unmatched	C: Quota	D: Quota + donor
			+0.001	+0.001	+0.001
0.0	0.0	0.5	[-0.007,+0.007]	[-0.007,+0.008]	[-0.006,+0.007]
			-0.000	-0.000	-0.000
0.0	0.0	1.5	[-0.003,+0.001]	[-0.003,+0.002]	[-0.003,+0.001]
			+0.024	-0.001	-0.002
0.0	1.5	0.5	[-0.005,+0.075]	[-0.012,+0.010]	[-0.012,+0.007]
			+0.002	+0.000	-0.000
0.0	1.5	1.5	[-0.005,+0.009]	[-0.005,+0.003]	[-0.003,+0.005]
			+0.028	+0.028	+0.031
1.5	0.0	0.5	[-0.028,+0.103]	[-0.023,+0.103]	[-0.015,+0.104]
			+0.043	+0.042	+0.042
1.5	0.0	1.5	[-0.005,+0.114]	[-0.006,+0.113]	[-0.009,+0.115]
			+0.027	+0.016	+0.025
1.5	1.5	0.5	[-0.051,+0.133]	[-0.059,+0.115]	[-0.039,+0.103]
			+0.040	+0.029	+0.048
1.5	1.5	1.5	[-0.029,+0.130]	[-0.054,+0.114]	[-0.008,+0.136]

Table 1: Synthetic mechanism check: subject-balanced Macro-F1 difference between 32 and 8 training subjects at fixed total 1,600 cells. Entries are means across 20 independent seeds with empirical 2.5th–97.5th seed percentiles, not confidence intervals of the mean. Condition B equals A in every run because support is already complete and the support floor does not alter selection. All eight parameter settings are reported.

D results are not included in this version; the existing class-matched tables must not be interpreted as that decomposition.

2.2 KNOWN-NULL AND INDEPENDENT-TARGET VALIDATION

Known generator parameters are not themselves a known learning effect. We therefore define the target explicitly as

$$\theta_C = \mathbb{E}[S_C(32, Q_8) - S_C(8, Q_8)], \quad (1)$$

where S is subject-balanced pooled six-subtype Macro-F1 on untouched test subjects, and Q_8 is the development-only eight-subject class-quota reference. The expectation integrates over generated cohorts and sampling. This overlap-quota policy target excludes gains from acquiring labels absent at $P = 8$; it is not the total practical value of recruiting more subjects.

We generate a hierarchy of three lineages with two sibling subtypes each in eight dimensions. Fixed class centers are $\mu_k = 2e_{\lfloor k/2 \rfloor} + (-1)^{k+1} se_{3+\lfloor k/2 \rfloor}$ for $k = 0, \dots, 5$ (zero-indexed axes). Features follow $x_{pi} = \mu_{y_{pi}} + b_p + u_{p,g(i)} + v_{h(p)} + \epsilon_{pi}$, with independent isotropic Gaussian effects: donor standard deviation a , ten-cell block standard deviation c , four-donor batch standard deviation b , and residual standard deviation 1. Within-subject correlation can thus arise at donor or cell-block level; these covariance components should not be interpreted as independent empirical estimates of biological heterogeneity. Subject label probabilities are softmax-normal with logit standard deviation 1.5, and the sixth subtype is available in only 10% of subjects. Each cohort contains 32 development and 40 test subjects, with 120 cells per subject. The training total is 480 at both $P = 8$ and $P = 32$; preprocessing and LR fitting follow the preceding synthetic probe. Train and test subjects and batches are disjoint. All tasks retain the fixed six-class scoring ontology.

The base setting is $(a, c, b, s) = (0, 0, 0, 0.8)$. Conditional on the label-only quota reference, training features in each class are iid with the same distribution for both subject budgets. Consequently, the C fitted-model distribution, and hence expected test score, is identical: $\theta_C = 0$ exactly. This argument does not require balanced natural class proportions or complete support. Additional settings change only a to 0.5 or 1.5; stress settings start from $a = 0.5$ and set $c = 0.7$, $s = 0.3$, or $b = 1$, respectively. In the batch setting, increased subject breadth also increases batch coverage; the target remains a policy contrast rather than a separate donor effect.

For every non-null setting we estimate θ_C using 512 independent reference cohorts, with a direct uniform-without-replacement index sampler separate from the evaluated priority-sampling implementation. Another 100 independent cohorts evaluate A/B/C using paired subject pools, cell priorities, and test sets. Reference and evaluation random-seed ranges are disjoint. We report reference

Monte Carlo standard errors (MCSE), estimator bias, and RMSE. Normal Monte Carlo intervals for bias combine evaluation-mean variance and independent reference-mean variance; these are not patient-bootstrap coverage measurements or multiplicity-adjusted tests.

Setting	Target (MCSE)	A: mean / RMSE	B: mean / RMSE	C: mean / RMSE
Known null	0 (exact)	+0.0202 / 0.0354	+0.0119 / 0.0249	+0.0004 / 0.0109
Mild donor shift	0.0110 (0.0009)	+0.0278 / 0.0341	+0.0221 / 0.0264	+0.0102 / 0.0191
Strong donor shift	0.0135 (0.0015)	+0.0332 / 0.0412	+0.0316 / 0.0392	+0.0168 / 0.0358
Correlated cells	0.0108 (0.0008)	+0.0301 / 0.0334	+0.0264 / 0.0290	+0.0102 / 0.0199
Weak subtypes	0.0049 (0.0010)	+0.0247 / 0.0361	+0.0235 / 0.0346	+0.0016 / 0.0222
Batch shift	0.0181 (0.0017)	+0.0318 / 0.0463	+0.0299 / 0.0438	+0.0192 / 0.0429

Table 2: Known-null and independent Monte Carlo validation of the overlap-quota target. A, B, and C denote unmatched, support-matched, and exact-quota sampling. Non-null targets use 512 independent reference cohorts; estimator means and RMSE use 100 separate evaluation cohorts. MCSE denotes reference Monte Carlo standard error. The target is the expected effect of the C allocation policy, not unrestricted subject recruitment. All six settings are reported; RMSE is measured relative to the exact null or estimated reference target.

Under the exact null, A and B yield mean effects +0.0202 and +0.0119, whereas C yields +0.0004 (bias interval $[-0.0017, 0.0026]$). The independent reference check is +0.00043 with MCSE 0.00053, consistent with the analytic null. Mild and strong donor shifts have reference effects +0.0110 and +0.0135; C estimates +0.0102 and +0.0168, respectively, rather than mechanically forcing both to zero. Higher shift variance does not guarantee a proportionally larger learning benefit. Across all six settings, C has lower RMSE against this target than A or B. Nevertheless, A/B differences can be legitimate benefits of their own allocation policies: their error relative to θ_C demonstrates estimand mismatch, not that all observed breadth gains are spurious. Non-null targets remain Monte Carlo approximations. This validation neither identifies real-cohort causal mediation nor establishes coverage of the joint uncertainty procedure used on real data.

3 REAL-COHORT SAMPLING AUDITS

3.1 SUPPORTING SUBJECT-CELL SAMPLING AUDIT

The following analyses address sampling-policy sensitivity, not the clinical validity of disease prediction. They preserve the completed cross-cohort evidence as support for the cell-to-patient evaluation.

Two allocation estimands. For a specified sampling policy π and learning procedure f , define

$$\Delta_\pi(T) = \mathbb{E}[S(f_{\pi,32,T}) - S(f_{\pi,8,T})], \quad (2)$$

where S is subject-balanced pooled macro-F1 and the expectation averages training-subset randomization under the fixed cohort and evaluation protocol. Natural sampling estimates a *total allocation effect*, including label distribution changes. The matched policy estimates a *label-standardized allocation effect* relative to the sampled $P = 8$ quota vector, averaged over these reference draws. It also imposes within-class donor round robin. These are different policy targets, not competing estimators of a uniquely true patient effect; neither isolates biological diversity from batch structure.

Stability is not uncertainty of the mean. The released joint stability interval is the central 95% range obtained by selecting one completed subset seed and bootstrapping paired test subjects. It retains between-subset variability rather than averaging resampled seeds. Its zero inclusion does not test $\Delta_\pi(T) = 0$, and its positive-replicate fraction is neither a posterior probability nor a significance test. Inference for the seed-averaged effect requires a separately constructed interval; the present main comparisons report stability, not that inferential claim.

3.2 FIXED TOTAL CELLS DO NOT ISOLATE SUBJECT BREADTH

We begin with the conventional label-agnostic design: outer-test subjects remain fixed while a constant number of training cells is reallocated from 8 to 32 subjects. Across APT, COMBAT, and

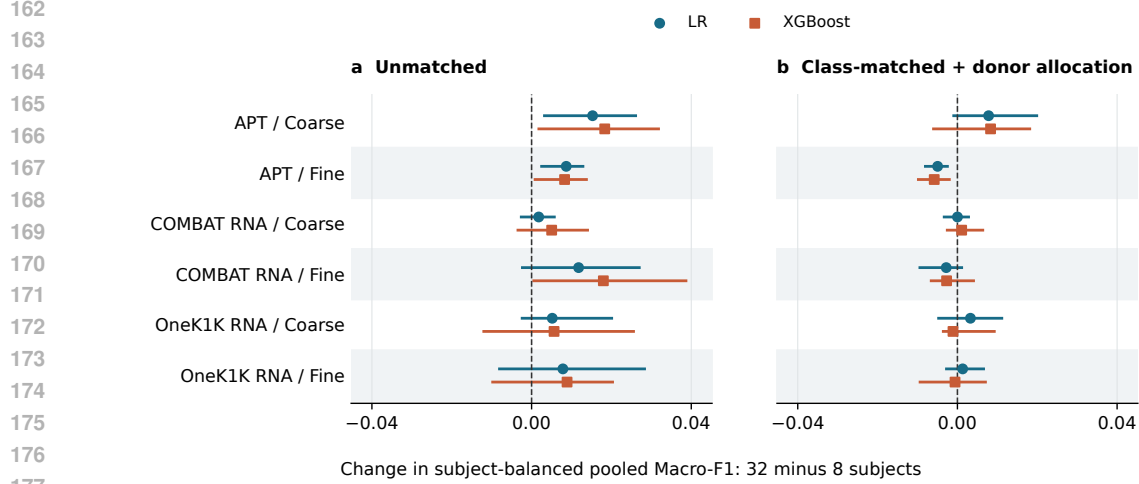


Figure 1: The apparent breadth advantage changes under the matched protocol. Both panels show the change in subject-balanced pooled OOF macro-F1 from 8 to 32 training subjects at $T = 6,400$, on identical axes. The right panel fixes class support and exact quotas and additionally uses within-class donor round robin; this is not a label-only intervention. Points are seed means; bars in both panels are empirical 2.5th–97.5th seed percentiles, not confidence intervals of the mean. Unmatched runs and matched LR use 20 seeds; matched XGBoost uses 10. APT and COMBAT fine point estimates change sign for both models. Cross-panel changes are descriptive, not paired attenuation tests. The matched joint stability intervals, which additionally resample held-out subjects, all span zero (Figure 3).

OneK1K, 37 of 38 model–resolution–budget comparisons favor greater subject breadth. The direction appears strikingly consistent across assays and cohorts.

It is not an isolated subject effect. Increasing the number of sampled subjects also changes which labels are observed, how the fixed cell budget is distributed among classes, and the resulting class proportions. For example, the mean number of observed fine classes rises with subject count in all three datasets, despite fixed total cells. Training-subset intervals are also much less stable than intervals that condition on the averaged training subsets and bootstrap only test subjects. The label-agnostic experiment therefore measures a total allocation effect, not a label-standardized effect. Conditioning on completed training subsets also omits sensitivity to alternative development samples.

3.3 BREADTH ESTIMATES DEPEND ON THE ALLOCATION POLICY

The matched comparison freezes the $P = 8$ observed class set and exact per-class quotas, then samples those same quotas from 8 and 32 subjects. Every joint replicate first selects one completed training-subset seed and then resamples paired outer-test subjects. This targets the residual effect of redistributing a fixed class-matched cell budget across more subjects. The resulting interval describes joint stability, not precision of the seed mean.

The matched policy yields a different pattern (Figure 1). At $T = 6,400$, LR/XGBoost effects are $+0.0078/+0.0083$ for APT coarse but $-0.0050/-0.0058$ for APT fine. COMBAT effects are approximately zero for coarse and $-0.0028/-0.0027$ for fine. OneK1K effects are small and change sign across models. All 12 joint stability intervals include zero. The APT and COMBAT fine point-estimate reversals are therefore not specific to a linear classifier, but their signs are not stable across joint resampling. This is a joint sampling-protocol contrast: the implementation fixes observed labels, class counts, and proportions and also uses within-class donor round robin. The attenuation cannot therefore be assigned solely to label distribution. An independently controlled synthetic check separates quota matching from donor allocation (Section 2.1); it does not replace this separation on the real cohorts. A complementary known-null validation yields mean effects $+0.0202$, $+0.0119$, and $+0.0004$ for unmatched, support-matched, and exact-quota sampling against an exactly zero overlap-quota target. Independent Monte Carlo references under donor shifts remain

positive and are not mechanically removed by matching (Section 2.2). This checks recovery of a specified allocation target, not interval coverage or a universal causal effect of subject breadth.

Beyond the common 32-subject grid, class-matched LR frontiers through 64 COMBAT and 128 OneK1K training donors yield adjacent mean effects between -0.00065 and $+0.00123$, with every joint stability interval spanning zero. These ranges do not test mean equivalence or a plateau. A post-hoc ± 0.01 sensitivity band contains 11 of 12 adjacent frontier stability intervals, but only 4 of 12 main $T = 6,400$ intervals and none of the four APT intervals (Section 3.8). These are pointwise interval-containment diagnostics, not equivalence tests or an optimal-allocation result. Complete unmatched grids, class-support audits, matched doublings, extended frontiers, and uncertainty decompositions appear in Section 3.4–3.7.

3.4 FAIR FIXED-TOTAL AND MATCHED-SCALE ANALYSIS

The supporting scaling design uses $P \in \{8, 16, 32\}$ and $C \in \{100, 200, 400, 800, 1600\}$, with 20 nested subset seeds for every outer fold. Because every retained patient has at least 1,929 cells, all finite caps are feasible without replacement. Disease-stratified patient orderings are nested across P , uniform cell permutations are nested across C , no subtype label enters sampling, and preprocessing is refit on each sampled training set. The fixed-total diagonals satisfy $PC \in \{3200, 6400, 12800\}$ exactly.

Model	Task	Total cells	$P = 8$	$P = 16$	$P = 32$
LR	Coarse	3,200	0.270 [0.259, 0.280]	0.277 [0.270, 0.285]	0.282 [0.279, 0.286]
LR	Coarse	6,400	0.277 [0.266, 0.290]	0.286 [0.278, 0.293]	0.292 [0.290, 0.295]
LR	Coarse	12,800	0.281 [0.270, 0.294]	0.291 [0.284, 0.300]	0.298 [0.296, 0.301]
LR	Fine	3,200	0.078 [0.073, 0.084]	0.082 [0.079, 0.087]	0.085 [0.083, 0.089]
LR	Fine	6,400	0.080 [0.077, 0.087]	0.085 [0.081, 0.089]	0.089 [0.087, 0.091]
LR	Fine	12,800	0.081 [0.076, 0.087]	0.087 [0.083, 0.090]	0.091 [0.089, 0.094]
XGBoost	Coarse	3,200	0.262 [0.250, 0.276]	0.272 [0.260, 0.284]	0.279 [0.274, 0.284]
XGBoost	Coarse	6,400	0.269 [0.258, 0.284]	0.279 [0.267, 0.289]	0.288 [0.285, 0.292]
XGBoost	Coarse	12,800	0.276 [0.267, 0.290]	0.289 [0.277, 0.298]	0.298 [0.296, 0.300]
XGBoost	Fine	3,200	0.082 [0.078, 0.088]	0.087 [0.081, 0.092]	0.089 [0.086, 0.092]
XGBoost	Fine	6,400	0.086 [0.081, 0.094]	0.091 [0.086, 0.097]	0.095 [0.090, 0.098]
XGBoost	Fine	12,800	0.092 [0.087, 0.098]	0.097 [0.092, 0.102]	0.098 [0.096, 0.101]

Table 3: Fixed-total training-cell scaling. Each row holds the total number of sampled training cells constant while reallocating them across independent patients. Entries are subject-balanced pooled, patient-disjoint OOF Macro-F1 means with empirical 95% intervals over 20 matched subset seeds. For each score, every held-out subject’s confusion matrix is normalized to unit mass before pooling; this is distinct from both cell-weighted pooling and mean within-subject Macro-F1. All outer-test cells are evaluated.

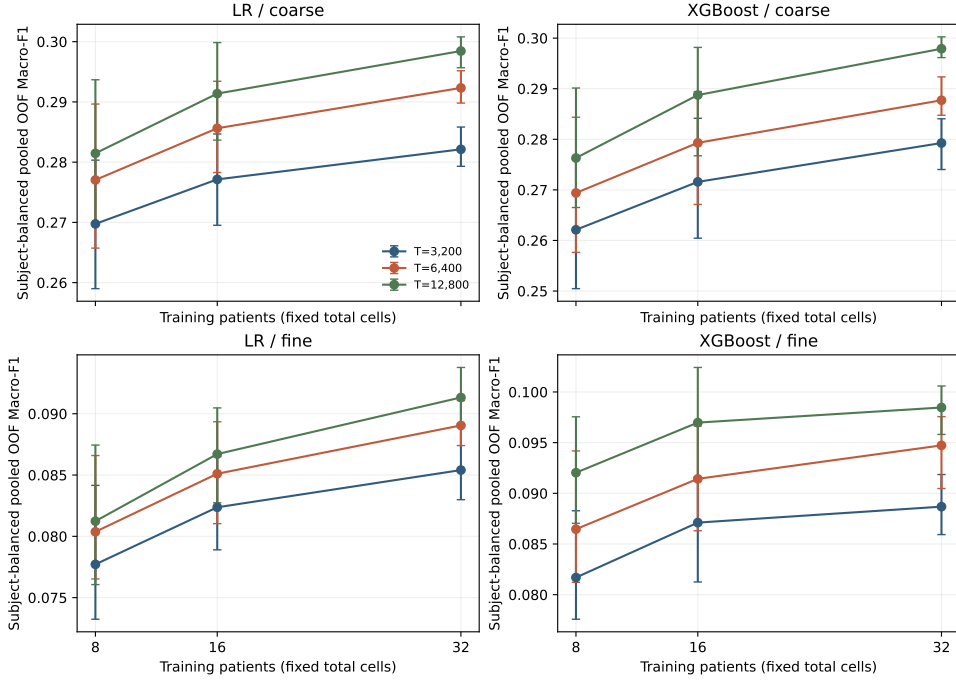


Figure 2: APT-only fixed-total response curves underlying the unmatched cross-cohort diagnostic in Table 4. Points are pooled OOF macro-F1 means over 20 matched subset seeds; bars are empirical 95% seed intervals. Patient-balanced and patient-bootstrap results are provided in the machine-readable artifacts.

Table 3 reports variability across subset seeds. Separately, paired patient-clustered bootstraps over the same 40 pooled OOF patients compare $P = 8$ with $P = 32$ after averaging seed-level sufficient statistics. All 12 intervals exclude zero; their lower endpoints range from 0.0002 to 0.0084 and upper endpoints from 0.0090 to 0.0314. These intervals quantify uncertainty over patients, whereas the empirical seed intervals quantify sensitivity to which development patients and cells are sampled.

Matched doubling effects average patient versus cell gains of 0.011/0.005 (LR coarse), 0.007/0.003 (LR fine), 0.016/0.008 (XGBoost coarse), and 0.009/0.006 (XGBoost fine). In the descriptive fold-fixed-effect response surface, the corresponding per-doubling β_P/β_C coefficients are 0.017/0.008, 0.008/0.004, 0.016/0.007, and 0.009/0.006. Complete pairwise effects, patient intervals, seed intervals, and response-surface uncertainty are stored in the machine-readable scaling artifacts. The coefficients are associations over this grid, not causal or universal scaling laws.

3.5 EXTERNAL COMBAT AND ONEK1K SCALING REPLICATIONS

We downloaded the author-released COMBAT CITE-seq object (COVID-19 Multi-omics Blood Atlas (COMBAT) Consortium, 2022) (836,148 cells and 124 donor identifiers; SHA-256 f628a15f25b9ca2f7cdefeab271fd9d007c8d5e47eb80c4b807d5f65e86ff53d). We removed cells without a resolved major or fine immune annotation and cells flagged as GEX-region doublets. Ten donors were frozen as a calibration-only set and excluded from every model fit and test fold. Among the remaining donors, requiring at least 1,600 eligible cells leaves 105 donors and 700,706 cells. We construct five disjoint folds of 21 test donors, stratified by the released sample-source field, and freeze these folds across all models, budgets, seeds, and modalities.

The RNA replication uses 293 protein-coding genes with highest variance on the 10 calibration donors after excluding mitochondrial and ribosomal genes. This matches the APT input dimension without selecting features on any evaluation donor. The ADT sensitivity uses all 192 released antibody features. Both inputs use the released five-class major annotation and 39-class fine an-

notation. They share the exact P , C , and fixed-total grids, 20 nested source-stratified donor/cell seeds, training-only standardization, LR configuration, and paired donor-bootstrap procedure used in APT-Bench. The RNA analysis additionally uses the frozen 400-tree XGBoost baseline. Every fit is evaluated on all cells of the 21 untouched test donors; the ADT analysis is an LR-only modality sensitivity rather than an additional model-comparison family. Dataset hashes, folds, features, sampling manifests, and per-donor sufficient statistics are included in the release artifacts.

OneK1K provides a second independent scRNA-seq replication in a much larger healthy PBMC cohort (Yazar et al., 2022). We use the current author-deposited CELLxGENE object (1,248,980 cells and 981 donor identifiers; SHA-256 14c5e9e70007a5e7dc047f7b29405773305e1a7b9791d0789af70649cf6f46ec), remove released doublet calls, and reserve one complete multiplexing pool for calibration. Requiring at least 800 eligible cells leaves 925 evaluation donors and 1,199,468 cells. Five folds are disjoint in both donor and multiplexing pool. We select 293 protein-coding, non-mitochondrial, non-ribosomal genes by variance on ten calibration-pool donors only, and use the released 30-class Azimuth-derived fine labels with a deterministic five-lineage mapping.

OneK1K uses the same models, 20 nested subset seeds, training-only standardization, untouched-test-donor evaluation, subject-balanced pooled metric, and 2,000 paired donor bootstraps. Its grid is $P \in \{8, 16, 32\}$ and $C \in \{100, 200, 400, 800\}$, yielding the two common exact budgets $T \in \{3200, 6400\}$. We prespecified the lower depth ceiling because only 116 of the 981 raw donors have at least 1,600 eligible cells; enforcing the full APT grid would discard most donors and condition the replication on high cell recovery. OneK1K tests scaling portability in a healthy cohort, not disease generalization or absolute accuracy across different feature spaces and label-generation procedures.

Dataset	Modality	Model/task	$T = 3,200$	$T = 6,400$	$T = 12,800$
APT	Aptamer	LR/Coarse	+0.012 [+0.005, +0.021]	+0.015 [+0.007, +0.024]	+0.017 [+0.009, +0.026]
APT	Aptamer	LR/Fine	+0.008 [+0.004, +0.011]	+0.009 [+0.005, +0.012]	+0.010 [+0.005, +0.013]
APT	Aptamer	XGBoost/Coarse	+0.017 [+0.008, +0.027]	+0.018 [+0.007, +0.030]	+0.022 [+0.009, +0.035]
APT	Aptamer	XGBoost/Fine	+0.007 [+0.002, +0.011]	+0.008 [+0.002, +0.013]	+0.006 [+0.000, +0.011]
COMBAT	RNA	LR/Coarse	+0.003 [+0.002, +0.004]	+0.002 [+0.001, +0.003]	+0.001 [+0.001, +0.002]
COMBAT	RNA	LR/Fine	+0.012 [+0.009, +0.013]	+0.012 [+0.010, +0.012]	+0.007 [+0.005, +0.009]
COMBAT	RNA	XGBoost/Coarse	+0.006 [+0.005, +0.007]	+0.005 [+0.004, +0.006]	+0.004 [+0.003, +0.005]
COMBAT	RNA	XGBoost/Fine	+0.016 [+0.013, +0.017]	+0.018 [+0.016, +0.019]	+0.017 [+0.014, +0.018]
COMBAT	ADT	LR/Coarse	+0.006 [+0.004, +0.008]	+0.009 [+0.006, +0.012]	+0.012 [+0.010, +0.016]
COMBAT	ADT	LR/Fine	+0.013 [+0.010, +0.014]	+0.011 [+0.010, +0.013]	+0.011 [+0.009, +0.013]
OneK1K	RNA	LR/Coarse	+0.004 [+0.004, +0.005]	+0.005 [+0.005, +0.006]	–
OneK1K	RNA	LR/Fine	+0.007 [+0.007, +0.009]	+0.008 [+0.007, +0.009]	–
OneK1K	RNA	XGBoost/Coarse	-0.003 [-0.004, -0.002]	+0.006 [+0.005, +0.006]	–
OneK1K	RNA	XGBoost/Fine	+0.006 [+0.005, +0.007]	+0.009 [+0.007, +0.009]	–

Table 4: Unmatched diagnostic: change in subject-balanced pooled, subject-disjoint OOF Macro-F1 when the same total training-cell budget is reallocated from 8 to 32 independent subjects without freezing the marginal training label distribution. Each held-out subject confusion matrix is normalized to unit mass before pooling. Brackets are 95% paired subject-bootstrap intervals over the fixed outer-test subjects. Positive values favor broader subject coverage but can also reflect changes in observed classes, per-class counts, and class proportions. COMBAT ADT is a prespecified modality sensitivity using LR only. OneK1K uses the two common exact budgets supported by at least 800 cells from nearly all donors.

3.6 LABEL-DISTRIBUTION AUDIT AND MATCHED CONTROL

The label-agnostic grid samples cells uniformly within subjects and therefore allows the full marginal label distribution to vary. Machine-readable audit tables report the observed class count, unseen-at-training class count, cell count, and donor count for every class at every fold, seed, and scaling point. The main text reports the common fixed-total diagonals, where fine-class coverage increases slightly with P despite fixed T ; this coverage statistic is only one observable component of the broader label-distribution effect.

The class-matched control fixes the marginal label distribution. The $P = 8$ subset defines the observed labels and a proportional integer quota T_k with a one-cell floor; capacity violations are deterministically redistributed. The same quota vector is then used at $P = 8, 16, 32$. Within each

class, a capacity-constrained round-robin sampler distributes cells across selected subjects before assigning a second cell from the same subject, so increasing P changes realized donor coverage rather than class counts. Sampling is without replacement, standardization is refit on each training subset, and all immutable outer-test cells are retained. All three runs complete 1,200/1,200 jobs and pass checks for exact total cells, identical paired class counts, non-overlapping test subjects, and contribution by every nominal training subject. These primary Logistic Regression runs use 20 matched subset seeds at both $T = 3,200$ and $T = 6,400$. To test whether the matched-policy direction is specific to a linear classifier, we repeat the identical control with the frozen XGBoost configuration at $T = 6,400$ using 10 matched seeds. Each dataset completes 300/300 additional jobs and passes the same checks; final production fits use the CPU histogram backend throughout to avoid mixing computational backends. It does not separately identify changes in disease-subtype composition, within-class allocation across subjects, or within-class subject heterogeneity; these remain bundled with the subject-breadth intervention.

Dataset	Model	Task	T	Classes (mean/total)	Donors/class (P : 8→32)	Δ SB-pooled M-F1	$\Pr(\Delta > 0)$
APT	LR	Coarse	3,200	5.0/5	7.9→28.5	+0.0062 [-0.0080, +0.0221]	0.778
APT	LR	Coarse	6,400	5.0/5	7.9→30.5	+0.0078 [-0.0070, +0.0265]	0.836
APT	LR	Fine	3,200	26.8/27	7.1→25.0	-0.0024 [-0.0077, +0.0029]	0.163
APT	LR	Fine	6,400	26.8/27	7.2→27.4	-0.0050 [-0.0105, +0.0009]	0.043
APT	XGBoost	Coarse	6,400	5.0/5	7.9→30.5	+0.0083 [-0.0130, +0.0265]	0.817
APT	XGBoost	Fine	6,400	26.8/27	7.2→27.3	-0.0058 [-0.0143, +0.0027]	0.110
COMBAT RNA	LR	Coarse	3,200	5.0/5	8.0→29.0	+0.0011 [-0.0036, +0.0084]	0.611
COMBAT RNA	LR	Coarse	6,400	5.0/5	8.0→31.3	+0.0000 [-0.0055, +0.0045]	0.547
COMBAT RNA	LR	Fine	3,200	39.0/39	7.2→22.5	-0.0007 [-0.0088, +0.0092]	0.439
COMBAT RNA	LR	Fine	6,400	39.0/39	7.3→26.6	-0.0028 [-0.0114, +0.0043]	0.272
COMBAT RNA	XGBoost	Coarse	6,400	5.0/5	8.0→31.2	+0.0011 [-0.0054, +0.0088]	0.522
COMBAT RNA	XGBoost	Fine	6,400	39.0/39	7.3→26.6	-0.0027 [-0.0096, +0.0066]	0.264
OneK1K RNA	LR	Coarse	3,200	5.0/5	7.8→27.7	+0.0025 [-0.0079, +0.0120]	0.669
OneK1K RNA	LR	Coarse	6,400	5.0/5	7.8→29.6	+0.0033 [-0.0059, +0.0128]	0.728
OneK1K RNA	LR	Fine	3,200	29.0/30	6.2→19.8	+0.0019 [-0.0080, +0.0105]	0.700
OneK1K RNA	LR	Fine	6,400	29.0/30	6.4→22.1	+0.0013 [-0.0058, +0.0089]	0.598
OneK1K RNA	XGBoost	Coarse	6,400	5.0/5	7.8→29.5	-0.0011 [-0.0061, +0.0139]	0.209
OneK1K RNA	XGBoost	Fine	6,400	28.9/30	6.4→22.0	-0.0006 [-0.0117, +0.0098]	0.549

Table 5: Class-matched fixed-total control. The outcome is subject-balanced pooled Macro-F1, computed by normalizing each held-out subject confusion matrix to unit mass before pooling. LR uses 20 seeds at both totals; the matched XGBoost robustness check uses 10 seeds at $T = 6,400$. Within each outer-fold/seed/task pair, the $P = 8$ subset freezes the observed classes and exact per-class cell quotas; the same quotas are used at $P = 16$ and $P = 32$. Brackets are joint 95% stability intervals obtained by resampling a training-subset seed and then paired outer-test subjects; the final column is the empirical fraction of joint replicates with a positive effect. These ranges are not confidence intervals for seed-averaged effects. Seed-only and subject-only intervals are released separately. Donors/class reports the mean number of subjects contributing cells to an observed class.

The primary stability diagnostic combines both variation sources. In each of 2,000 replicates, we sample one completed training-subset seed uniformly, resample outer-test subjects with replacement, use the same subject draw for $P = 8$ and $P = 32$, and recompute the paired difference. All 18 joint 95% stability intervals include zero, and the table reports the empirical positive-replicate fraction. For decomposition, empirical seed-only intervals include zero in 16 of 18 rows; the exceptions are the two negative APT fine results at $T = 6,400$. Subject-only intervals condition on the seed-averaged prediction matrix and can exclude zero even when the joint stability interval does not. The constructions answer different questions: the joint range retains variation between fitted training subsets rather than estimating uncertainty of their average. Neither zero inclusion nor the positive-replicate fraction is a test of the mean effect. These ranges also lack a demonstrated coverage guarantee for a future training cohort; they are conditional resampling diagnostics.

3.7 EXTENDED EXTERNAL SUBJECT FRONTIER

The common $P \leq 32$ grid is imposed by the 40-patient APT cohort, not by the capacity of the external datasets. We therefore extend the class-matched LR control to $P = 64$ for COMBAT and to $P \in \{64, 128\}$ for OneK1K at $T \in \{3,200, 6,400\}$. The original $P = 8$ subset freezes the class set and integer per-class quotas for every fold, seed, task, and total; only the number of selected subjects allowed to contribute those cells changes. Each new point uses 10 nested subset seeds,

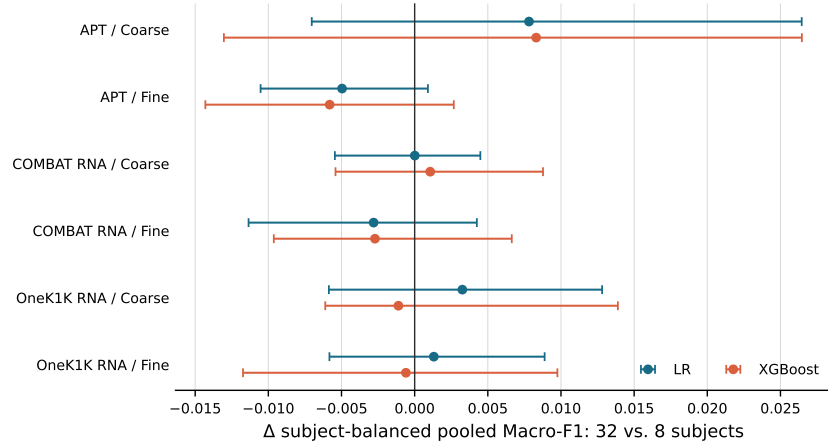


Figure 3: Joint stability for the completed class-matched protocol at $T = 6,400$. Points are mean changes in subject-balanced pooled OOF macro-F1 from 8 to 32 subjects. The 95% joint stability intervals select one training-subset seed and bootstrap paired held-out subjects; all 12 span zero. Unlike the seed-only bars in Figure 1, these intervals include test-subject variability. They are not confidence intervals for the seed-averaged effect; zero inclusion does not test a zero mean or establish equivalence.

all immutable outer-test subjects, and the same joint resampling of training-subset seed and held-out subjects used in the primary matched control. This extension is an external-cohort allocation sensitivity rather than a claim that its largest tested P is globally optimal.

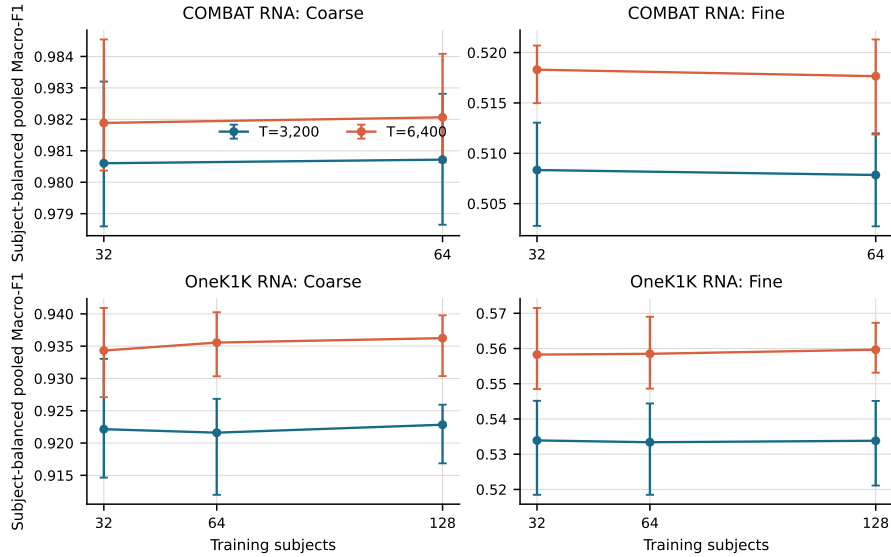


Figure 4: Class-matched LR frontier beyond 32 training subjects. Points are mean subject-balanced pooled OOF macro-F1 over 10 matched training-subset seeds; bars are empirical 95% seed intervals and therefore visualize subset sensitivity, not uncertainty of the mean. The paired joint stability intervals are reported in Table 6.

Dataset	Task	T	Subjects	Δ SB-pooled M-F1	$\Pr(\Delta > 0)$
COMBAT RNA	Coarse	3,200	32→64	+0.0001 [-0.0020, +0.0025]	0.535
COMBAT RNA	Coarse	6,400	32→64	+0.0002 [-0.0014, +0.0016]	0.618
COMBAT RNA	Fine	3,200	32→64	-0.0005 [-0.0060, +0.0026]	0.498
COMBAT RNA	Fine	6,400	32→64	-0.0006 [-0.0060, +0.0040]	0.450
OneK1K RNA	Coarse	3,200	32→64	-0.0005 [-0.0111, +0.0052]	0.553
OneK1K RNA	Coarse	6,400	32→64	+0.0012 [-0.0046, +0.0046]	0.766
OneK1K RNA	Fine	3,200	32→64	-0.0005 [-0.0060, +0.0053]	0.418
OneK1K RNA	Fine	6,400	32→64	+0.0002 [-0.0096, +0.0063]	0.595
OneK1K RNA	Coarse	3,200	64→128	+0.0012 [-0.0023, +0.0068]	0.642
OneK1K RNA	Coarse	6,400	64→128	+0.0007 [-0.0042, +0.0043]	0.657
OneK1K RNA	Fine	3,200	64→128	+0.0004 [-0.0039, +0.0050]	0.542
OneK1K RNA	Fine	6,400	64→128	+0.0012 [-0.0066, +0.0068]	0.617

Table 6: Extended class-matched subject frontier for Logistic Regression. Each comparison holds the total training-cell budget, observed class set, and exact per-class cell quotas fixed while increasing the number of contributing training subjects. Values are paired mean changes; brackets are joint 95% stability intervals from 2,000 replicates that sample a matched training-subset seed and then bootstrap held-out subjects. The final column is the empirical fraction of positive replicates, not a posterior probability. These ranges are not confidence intervals for seed-averaged effects.

All 12 adjacent frontier effects are small: their means range from -0.00065 to $+0.00123$, and every joint stability interval includes zero. The four direct OneK1K $32 \rightarrow 128$ contrasts likewise span zero under joint resampling. This does not establish a plateau or exclude practically meaningful changes. The explicit margin sensitivity below assesses interval width separately from zero inclusion. Because the intervention remains tied to these tasks, budgets, features, and LR configuration, it does not locate a universal saturation point or optimal allocation.

3.8 POST-HOC STABILITY-BAND SENSITIVITY

Failure to detect a difference is not evidence of equivalence (Lakens, 2017). After inspecting the existing results, we add an *exploratory* smallest-effect sensitivity bound $\epsilon = 0.01$ absolute subject-balanced pooled Macro-F1 (one percentage point), with 0.005 and 0.02 as sensitivity values. This is a reporting tolerance, not a validated clinical threshold, and was not prespecified for the completed experiments. It must not be conflated with the separate compact-panel tolerance $\delta = 0.02$.

For each released joint 95% stability interval $[L, U]$, we report zero inclusion ($L \leq 0 \leq U$) and strict containment ($-\epsilon < L$ and $U < \epsilon$) separately. Containment describes concentration of this conditional resampling distribution inside the chosen band, not equivalence of the mean allocation effects. Intervals with $L \leq -\epsilon$ and $U \geq \epsilon$ span changes of the stated magnitude in both directions. These diagnostics neither quantify statistical power nor provide a calibrated future-cohort guarantee.

At $T = 6,400$, only 4 of 12 main comparison intervals meet the ± 0.01 containment criterion; none of the four APT intervals does. The APT coarse XGBoost interval spans both meaningful directions. Among adjacent external LR frontier comparisons, 11 of 12 intervals meet this criterion; the exception is OneK1K coarse $32 \rightarrow 64$ at $T = 3,200$. With the tighter ± 0.005 margin, no main interval and only 4 of 12 adjacent frontier intervals are contained. Descriptive stability is therefore margin-dependent; these counts do not establish that the sampling policies are equivalent or subjects have no value.

We do not report formal TOST p -values. Conventional level-0.05 TOST is linked to a corresponding 90% confidence interval, whereas this analysis retains the released joint 95% stability intervals. More fundamentally, those intervals sample one completed training-subset seed and then test subjects, combining between-fit variability with test uncertainty rather than estimating only a seed-averaged mean effect. Containment is therefore an exploratory joint-interval screen, not a calibrated confirmatory equivalence test. Results are pointwise, without a simultaneous equivalence guarantee across rows, and do not establish a continuous plateau between or beyond the tested budgets. This audit concerns the completed legacy controls, not the ongoing A–D GPU rerun.

Comparison	Dataset	Task	n	$\pm .005$	$\pm .010$	$\pm .020$	Both signs*
8-32	APT	coarse	2	0	0	0	1
8-32	APT	fine	2	0	0	2	0
8-32	COMBAT	coarse	2	0	2	2	0
8-32	COMBAT	fine	2	0	1	2	0
8-32	OneK1K	coarse	2	0	0	2	0
8-32	OneK1K	fine	2	0	1	2	0
Extended LR	COMBAT	coarse	2	2	2	2	0
Extended LR	COMBAT	fine	2	0	2	2	0
Extended LR	OneK1K	coarse	4	2	3	4	0
Extended LR	OneK1K	fine	4	0	4	4	0

Table 7: Post-hoc stability-band sensitivity: number of released joint 95% stability intervals strictly inside each stated margin. The 8-32 rows use $T = 6,400$ and both LR and XGBoost; extended rows use LR at both totals for adjacent subject-budget comparisons. *Intervals reaching both -0.01 and $+0.01$, indicating that the resampling range spans changes of this magnitude in both directions. Counts are pointwise diagnostics, not mean-effect confidence intervals, formal TOST decisions, or evidence of equivalence.

Model	Task	$C = 100$	$C = 500$	$C = 2000$	$C = \text{all}$
Logistic Regression, $P = 8$	Coarse	0.255 [0.247, 0.265]	0.274 [0.263, 0.284]	0.283 [0.269, 0.297]	0.282 [0.269, 0.293]
Logistic Regression, $P = 16$	Coarse	0.266 [0.261, 0.270]	0.290 [0.284, 0.297]	0.288 [0.282, 0.295]	0.290 [0.280, 0.300]
Logistic Regression, $P = 24$	Coarse	0.276 [0.269, 0.280]	0.296 [0.288, 0.304]	0.288 [0.282, 0.295]	0.296 [0.289, 0.302]
Logistic Regression, $P = 32$	Coarse	0.283 [0.278, 0.288]	0.299 [0.296, 0.302]	0.287 [0.284, 0.288]	0.299 [0.284, 0.315]
Logistic Regression, $P = 8$	Fine	0.074 [0.068, 0.081]	0.084 [0.079, 0.088]	0.087 [0.082, 0.092]	0.101 [0.091, 0.109]
Logistic Regression, $P = 16$	Fine	0.081 [0.078, 0.085]	0.090 [0.088, 0.094]	0.095 [0.090, 0.099]	0.116 [0.110, 0.125]
Logistic Regression, $P = 24$	Fine	0.086 [0.083, 0.089]	0.093 [0.089, 0.097]	0.100 [0.095, 0.105]	0.125 [0.118, 0.131]
Logistic Regression, $P = 32$	Fine	0.089 [0.086, 0.092]	0.095 [0.093, 0.097]	0.105 [0.102, 0.108]	0.132 [0.112, 0.146]
XGBoost, $P = 8$	Coarse	0.246 [0.234, 0.263]	0.262 [0.250, 0.278]	0.277 [0.262, 0.293]	0.300 [0.283, 0.321]
XGBoost, $P = 16$	Coarse	0.260 [0.252, 0.272]	0.278 [0.268, 0.288]	0.295 [0.283, 0.306]	0.313 [0.298, 0.330]
XGBoost, $P = 24$	Coarse	0.270 [0.264, 0.276]	0.289 [0.284, 0.295]	0.305 [0.297, 0.312]	0.319 [0.311, 0.328]
XGBoost, $P = 32$	Coarse	0.275 [0.270, 0.280]	0.296 [0.293, 0.298]	0.310 [0.308, 0.312]	0.322 [0.300, 0.345]
XGBoost, $P = 8$	Fine	0.073 [0.068, 0.079]	0.085 [0.080, 0.091]	0.096 [0.090, 0.103]	0.118 [0.110, 0.126]
XGBoost, $P = 16$	Fine	0.082 [0.077, 0.085]	0.096 [0.092, 0.101]	0.110 [0.103, 0.120]	0.132 [0.125, 0.140]
XGBoost, $P = 24$	Fine	0.087 [0.081, 0.092]	0.100 [0.095, 0.104]	0.115 [0.110, 0.121]	0.138 [0.132, 0.144]
XGBoost, $P = 32$	Fine	0.090 [0.087, 0.093]	0.102 [0.100, 0.105]	0.118 [0.117, 0.121]	0.143 [0.127, 0.159]

Table 8: Patient-cell training scaling. Entries are pooled OOF Macro-F1 means with empirical 2.5th-97.5th seed intervals; the deterministic $P = 32, C = \text{all}$ endpoint instead uses a patient-clustered bootstrap interval. Seed intervals are not patient confidence intervals.

3.9 SECONDARY DESCRIPTIVE PATIENT-CELL SCALING SURFACE

This historical 4×4 surface fixes eight outer-test patients and varies only the training set. For each outer fold and each of 20 seeds, we generate one approximately proportional disease-stratified ordering of the 32 development patients, giving nested sets $S_8 \subset S_{16} \subset S_{24} \subset S_{32}$. Each selected patient receives one uniform random cell ordering, giving nested prefixes of 100, 500, and 2,000 cells and the complete available set. Patients with fewer than a requested cap contribute all available cells. No cell is selected using its subtype, and every model is evaluated on all cells from the untouched test patients. The full-data endpoint is fitted once per fold rather than duplicated across seeds. All 6,020 model-task jobs completed, and the deterministic endpoint exactly reproduces the frozen pooled OOF baselines within absolute tolerance 5×10^{-4} . Because $P=8 \rightarrow 32$ is a fourfold change whereas $C=100 \rightarrow \text{all}$ is a larger and patient-dependent change, this surface is descriptive only. All direct comparisons between the two axes use the fixed-total and matched-doubling analysis in the main paper.

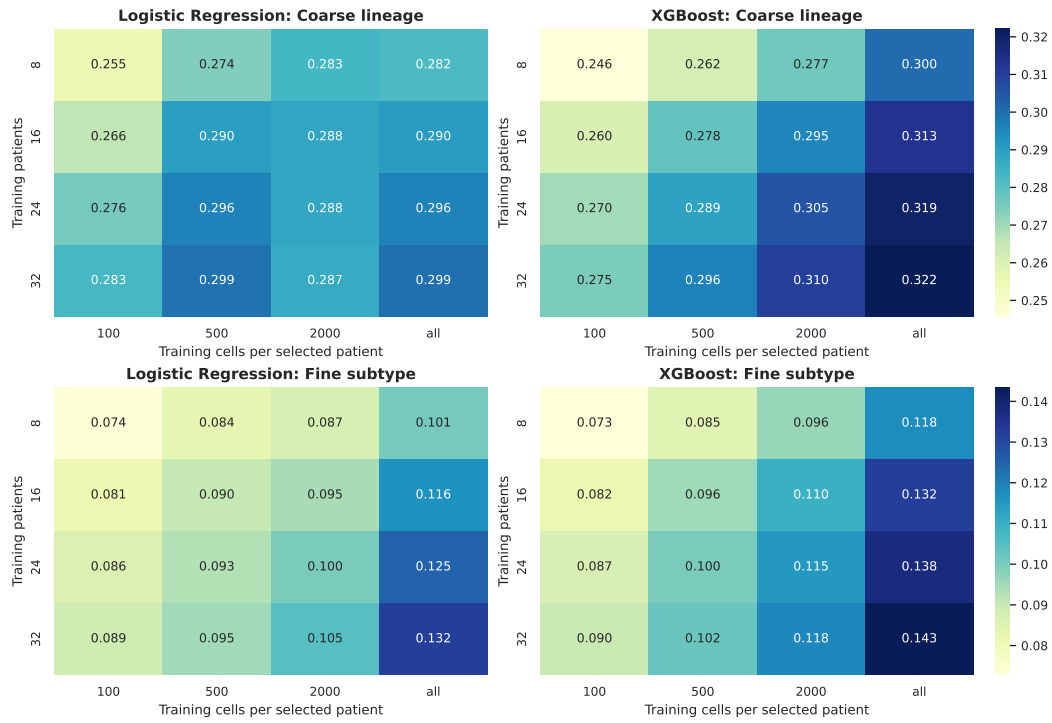


Figure 5: Cell-weighted pooled patient-disjoint Macro-F1 over the complete training-patient and per-patient training-cell grid. Values average 20 nested subset seeds except for the deterministic full-data endpoint. Color scales are shared across models within each task.

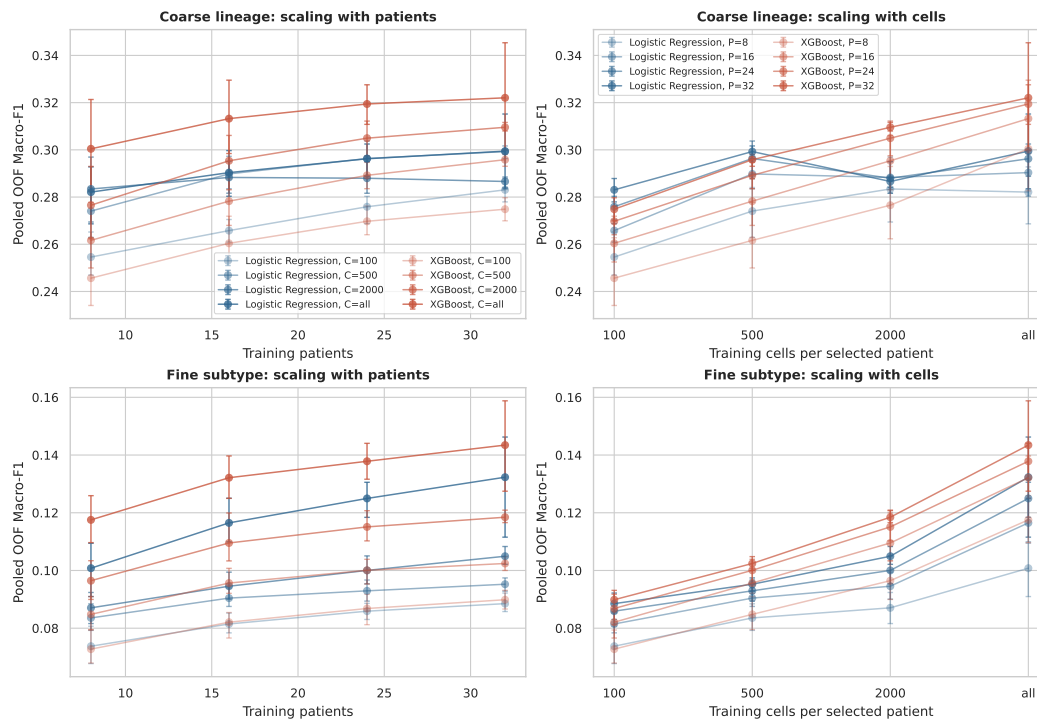


Figure 6: Patient-budget curves at each fixed per-patient cell cap. Intervals show subset-resampling variability; they do not represent uncertainty over a new patient population. Non-monotonic local changes are retained rather than smoothed.

The surface is cohort- and range-specific. We do not compare its unequal endpoints or use it to establish a patient- or cell-dominant scaling law, and it cannot estimate the benefit of patients drawn from new sites or acquisition batches.

REFERENCES

- COVID-19 Multi-omics Blood ATLAS (COMBAT) Consortium. A blood atlas of COVID-19 defines hallmarks of disease severity and specificity. *Cell*, 185(5):916–938.e58, 2022. doi: 10.1016/j.cell.2022.01.012.
- Daniel Lakens. Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, 8(4):355–362, 2017. doi: 10.1177/1948550617697177.
- Seyhan Yazar, Jose Alquicira-Hernandez, Kristof Wing, et al. Single-cell eqtl mapping identifies cell type-specific genetic control of autoimmune disease. *Science*, 376(6589):eabf3041, 2022. doi: 10.1126/science.abf3041.